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APPLIED PHYSICS LABORATORY

December 2025

INDICATORS OF BEHAVIOR OPERATIONAL PILOT

Summary Report from Pilot with HudsonAlpha Institute for Biotechnology

Version 1.0

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AOS-25-1688

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Portions of this document were edited with Generative AI (ChatGPT 5) to improve readability. All AI developed content was thoroughly reviewed by human subject matter experts for accuracy.

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1. EXECUTIVE SUMMARY

The Indicators of Behavior (IOB) Operational Pilot was conducted by the Johns Hopkins University Applied Physics Laboratory (APL) in partnership with the Cybersecurity and Infrastructure Security Agency (CISA), the Open Cybersecurity Alliance (OCA), and the HudsonAlpha Institute for Biotechnology. The purpose of the pilot was to determine whether behavior-based cyber threat intelligence could improve detection effectiveness and reduce analytic workload within a functioning Security Operations Center (SOC).

The pilot evaluated the integration of IOB bundles, expressed in Structured Threat Information eXpression (STIX) format, into HudsonAlpha's production environment. These bundles contained behaviors, detection analytics, and correlation workflows designed to identify coordinated adversary activity. The methodology combined threat emulation, laboratory validation, and an approximately three-month production run, beginning August 19 and concluding December 2, 2025.

Two use cases structured the technical evaluation. Use Case 1 focused on identifying multi-stage data theft activity observable through network telemetry. Use Case 2 focused on detecting adversary behaviors associated with manipulation of genomic research data within a laboratory setting. Both use cases were grounded in realistic threats to biotechnology operations and addressed security concerns that are common across the bioeconomy.

Across the full pilot period, the deployed analytics operated continuously and produced accurate, timely results. Thousands of individual behaviors were observed, and hundreds of partial correlations occurred, yet none of these generated operator notifications. Only fully correlated behavior sequences, all originating from planned adversary-emulation activities, resulted in escalations. Alerts were generated within five minutes after the final observed behavior. No false positives or erroneous notifications occurred at any point.

These outcomes demonstrate that IOB-based analytics can significantly improve the precision and efficiency of threat detection within an operational SOC. The pilot confirmed that behavior-driven correlation provides earlier and more reliable identification of coordinated adversary activity than traditional indicator-centric methods. The results also show that IOBs can be adopted in production environments without disrupting ongoing operations and with clear benefits to analyst workload and detection accuracy.

2. INTRODUCTION

IOBs represent a next-generation approach to cyber threat intelligence (CTI) that extends beyond traditional Indicators of Compromise (IOC). Whereas IOCs capture discrete artifacts left behind by adversaries, such as file hashes, IP addresses, or domains, IOBs codify higher-level behavioral patterns that describe how adversaries operate across the cyber kill chain. By focusing on repeatable adversary behaviors, IOBs enable earlier detection, support broader analytic reuse, and improve resilience against simple adversary changes intended to bypass defenses.

2.1 Background and Motivation

Despite these advantages, widespread adoption of new analytic standards such as IOBs typically lags behind their development. Many organizations are hesitant to integrate unproven technologies into live operational workflows without evidence of measurable benefit or minimal disruption to existing tools and processes. To overcome that barrier, it is necessary to demonstrate IOB effectiveness under realistic operational conditions, where existing tools, data sources, and workflows are already in active use. Controlled laboratory experiments are useful for early development, but only an operational pilot conducted in a production environment can validate the maturity, interoperability, and practical value of the technology.

To meet that need, APL, in partnership with CISA, OCA, and the HudsonAlpha Institute for Biotechnology, established an operational pilot to evaluate how IOBs can be translated into practical tools for security operations. The pilot provides a structured opportunity to validate the IOB model through direct collaboration with an operational SOC partner in a representative production environment.

2.2 Purpose of the Operational Pilot

The HudsonAlpha IOB Operational Pilot is designed to demonstrate, test, and evaluate the operational utility of IOBs in a functioning security operations setting. The effort seeks to move IOBs from theoretical promise to proven practice, bridging the gap between research and operational use.

Specifically, the pilot:

- Evaluated how IOB bundles, expressed in STIX format, can be integrated into a partner's SOC environment.
- Assessed the effectiveness of IOB-driven analytics in Security Information and Event Management (SIEM) platforms and in Security Orchestration, Automation, and Response (SOAR) workflows.
- Developed playbooks and metrics that quantify IOB performance, correlation efficiency, and operational value.
- Gathered partner feedback to guide refinements in IOB representation, tool interoperability, and usability for defenders.

By grounding the pilot in the operational realities of a biotechnology research campus, the pilot aims to produce measurable outcomes that inform CISA's broader strategy for technology transition and adoption across the bioeconomy and other U.S. critical infrastructure sectors.

2.3 Partner Organizations

The pilot represents a collaborative partnership among government, research, industry, and operational entities committed to advancing cyber defense capabilities and accelerating adoption of new analytic standards.

- Johns Hopkins University Applied Physics Laboratory: Serves as the lead technical organization and a designated University Affiliated Research Center (UARC) for the U.S. Government. APL is responsible for IOB bundle creation, emulation of the partner environment, integration of analytic tools, workflow development, and analysis of pilot data.
- HudsonAlpha Institute for Biotechnology: Acts as the operational pilot partner, providing SOC resources, architectural documentation, and log sources necessary for accurate replication of its environment within APL's integration infrastructure. HudsonAlpha operators participate in pilot evaluation activities and provide feedback on analytic performance, usability, and operational value.
- Cybersecurity and Infrastructure Security Agency: Serves as the federal sponsor, providing oversight of pilot progress and leveraging findings to inform broader IOB adoption across the bioeconomy and other sectors of national critical infrastructure.
- Open Cybersecurity Alliance: Provides the open standards foundation supporting the IOB framework. OCA is an international consortium of industry, government, and research organizations that promotes interoperability among cybersecurity tools through shared data models, specifications, and code libraries. The IOB framework and supporting pilot work directly contribute to OCA's mission of enabling vendor-neutral information exchange and automated defense.

This partnership model highlights the importance of linking federally sponsored research organizations with operational defenders and open standards bodies to demonstrate and transition emerging cybersecurity capabilities into production use.

2.4 Scope and Objectives

The IOB Operational Pilot was structured to bridge research and practice by designing and validating analytic capabilities within a controlled integration environment at APL before transitioning them into live operations on the HudsonAlpha Institute for Biotechnology network.

The APL integration environment provides a virtualized infrastructure that supports cybersecurity research and prototype development. Within this environment, APL replicated key elements of the HudsonAlpha architecture, configured representative tools, and validated IOB integrations across SIEM, SOAR, and endpoint platforms. This process allowed analytic workflows and playbooks to be developed, tested, and refined prior to deployment.

Following integration and internal validation, the pilot capabilities were transitioned to HudsonAlpha's operational SOC environment. The HudsonAlpha campus hosts high-throughput gene-sequencing platforms that generate large FASTQ/FASTA data sets, which are transferred across the shared network for downstream analysis. The live pilot focused on evaluating IOBs in

production workflows, analyzing their effectiveness in detecting and correlating adversary behaviors, and measuring operational performance.

APL provided technical support throughout the pilot, including bundle translation, analytic tuning, and data analysis. HudsonAlpha contributed operational requirements, environment configuration, and real-world event data. Together, the pilot team identified technical, procedural, and operational factors that influence IOB adoption and developed recommendations for scaling IOB use across other critical infrastructure sectors.

2.5 Structure of the Report

This report documents the planning, execution, and evaluation of the IOB Operational Pilot conducted jointly by APL, HudsonAlpha, and CISA. The report is organized to follow the pilot lifecycle and provide sufficient technical and operational detail for replication, transition, and future scaling.

The pilot was conducted in three distinct phases:

- Discovery Phase: Focused on identifying suitable pilot partners, defining the operational scope, and establishing the initial requirements for technology integration and evaluation.
- Design Phase: Involved configuration and integration of IOB-related capabilities within the APL test environment, replication of the partner's operational architecture, and transition planning for deployment within the HudsonAlpha environment.
- Execution Phase: Comprised the active use of IOB technologies within the pilot environment, collection of analytic and operational data, performance measurement, and assessment of results through partner feedback and technical analysis.

Subsequent sections of this report correspond to these phases and include detailed descriptions of pilot planning, technical design, execution activities, metrics, findings, and recommendations.

3. DISCOVERY PHASE SUMMARY

The Discovery Phase established the foundation for the IOB pilot through partner selection, technical alignment, and early use-case formulation. During this phase, APL identified HudsonAlpha Institute for Biotechnology as the operational collaborator, defined communication and coordination practices, characterized the production and laboratory environments, and selected representative technologies to support pilot execution. These activities ensured that subsequent pilot phases were grounded in accurate operational understanding and measurable, mission-relevant objectives.

3.1 Partner Selection

HudsonAlpha Institute for Biotechnology was selected as the operational partner for the IOB pilot following coordination meetings with CISA in late 2024. APL sought a collaborator capable of agile engagement and open technical exchange. HudsonAlpha's operational model of supporting a diverse set of biotechnology research programs and tenant organizations aligned with the pilot's goal of evaluating IOB applicability to emerging areas of U.S. Critical Infrastructure, particularly biotechnology and healthcare. CISA's interest in healthcare-adjacent use cases further reinforced the selection.

3.2 Kickoff and Initial Coordination

The pilot effort formally began with a kickoff meeting on December 13, 2024, bringing together APL and HudsonAlpha leadership to establish objectives, define potential use cases, and set communication protocols. Attendees included APL technical leads and HudsonAlpha's information technology and cybersecurity leadership. The kickoff established a collaborative structure consisting of biweekly strategic meetings, weekly technical exchanges, and ongoing coordination through email, Slack, and periodic site visits. Both organizations agreed to replicate representative aspects of the HudsonAlpha environment within APL's integration testbed and to focus the pilot on developing measurable, reusable security capabilities with relevance to the broader genomics community.

3.3 Site Visit and Environmental Characterization

A site visit in January 2025 enabled a detailed assessment of HudsonAlpha's operational and research environments. The visit included interviews with IT, security, and research personnel; review of architecture documentation and policies; and examination of security tooling and data workflows. HudsonAlpha's SOC provides network monitoring and defensive services for the campus and its biotechnology startups, functioning as both a managed-service and Internet-service provider. While the SOC maintains broad visibility into network activity, endpoint telemetry across independently managed tenant systems is limited. These characteristics informed the need for complementary use cases addressing both network-only and host-integrated perspectives.

3.4 Use Case Identification

Findings from the site visit guided the selection of two pilot use cases.

- Use Case 1: Focused on network-based data-exfiltration behaviors observable within HudsonAlpha's shared infrastructure. These behaviors include the movement of raw sequencing files produced by the institute's gene-sequencers, a critical data asset for the biotechnology tenant.
- Use Case 2: Examined endpoint- and integrity-focused adversary behaviors within a controlled research laboratory, demonstrating IOB adaptability beyond network telemetry.

Together, these use cases spanned the spectrum of cybersecurity challenges in biotechnology, from protecting multi-tenant network environments to safeguarding the integrity of genomic research data.

3.5 Technology Selection for Pilot Execution

To evaluate IOB effectiveness across varied data and operational domains, APL identified representative technologies for deployment in each pilot environment. The production-oriented use case incorporated a network monitoring capability, a SOAR platform, a SIEM platform, and an operator notification mechanism. The lab-oriented use case extended this stack with endpoint telemetry and related analytic capabilities. To prevent operators from receiving notifications unrelated to their production environment, APL did not implement an operator notification mechanism for the lab-oriented use case.

These technologies were selected to reflect capabilities common to research and healthcare organizations while maintaining interoperability across both open-source and commercial ecosystems. The resulting architecture provided a flexible foundation for translating, correlating, and visualizing IOB-driven analytics within the partner's operational context.

3.6 Transition to Design Phase

The Discovery Phase established the organizational relationships, technical baselines, and operational understanding necessary to inform the next stage of the pilot. Through direct engagement with HudsonAlpha and alignment with CISA priorities, APL validated the biotechnology sector as a relevant and challenging environment for IOB research. The findings from this phase, particularly the partner's network and laboratory characteristics, candidate use cases, and supporting technologies formed the basis for the detailed design activities described in the following section.

4. DESIGN PHASE SUMMARY

The Design Phase focused on translating the objectives established during the discovery activities into a reproducible technical framework capable of supporting IOB development, integration, and evaluation. This phase centered on recreating the HudsonAlpha operational environment within APL's integration infrastructure, defining a reference architecture to guide implementation, and developing the specific use cases, threat narratives, and behaviors underpinning the pilot. Through progressive design and iterative testing, APL ensured that data sources, analytic tools, and

detection logic were configured to enable consistent observation, emulation, and measurement of IOB performance prior to migration into the HudsonAlpha production environment.

4.1 Architecture Design

Prior to pilot deployment, APL constructed a representative architecture within its integration environment to serve as a controlled platform for technical development, validation, and threat emulation. This environment provided a means to replicate the operational structure of the HudsonAlpha campus while maintaining flexibility to iteratively integrate and test analytic and automation capabilities. By modeling the core components of the production network, including identity services, logging infrastructure, and operational enclaves, APL was able to simulate realistic data flows, exercise threat behaviors, and evaluate the performance of IOB analytics before introducing them into live systems.

The integration environment also enabled the configuration and validation of supporting technologies required to process, visualize, and execute IOBs. Within this environment, APL conducted threat emulation activities to identify adversary behaviors, integrated sensor technologies and telemetry collection mechanisms, and developed automation workflows to enable end-to-end testing of the detection analytics and correlation logic found within shared IOB bundles. The resulting architecture provided a repeatable baseline for analytic development and a controlled bridge between laboratory experimentation and production deployment.

4.1.1 Generalized Architecture

The representative architecture recreated within APL's integration environment mirrors the primary components of the HudsonAlpha production network, providing a realistic foundation for testing IOB workflows and emulation activities. The simulated environment includes a main campus enclave containing core enterprise services and operational functions typical of a research campus. Within this enclave, a SIEM platform aggregates network, endpoint, and system telemetry; SOC and Network Operations Center (NOC) operator workstations support analytic and administrative activities; and domain controllers manage authentication and access control services.

To capture the unique operational characteristics of the HudsonAlpha ecosystem, the architecture also incorporates a notional genomics tenant VLAN, representing one of the biotechnology organizations operating on the HudsonAlpha campus. This tenant enclave models scientific workloads, data exchanges, and research traffic that traverse the shared campus network. The inclusion of this VLAN enables the evaluation of cross-enclave visibility and the observation of security-relevant behaviors that may occur between enterprise and tenant environments. Figure 1 depicts the representative architecture used as the foundation for design and testing activities. The VLAN houses the campus's gene-sequencing pipelines, where gene sequencers continuously stream FASTQ data to storage clusters, providing realistic traffic for the pilot.

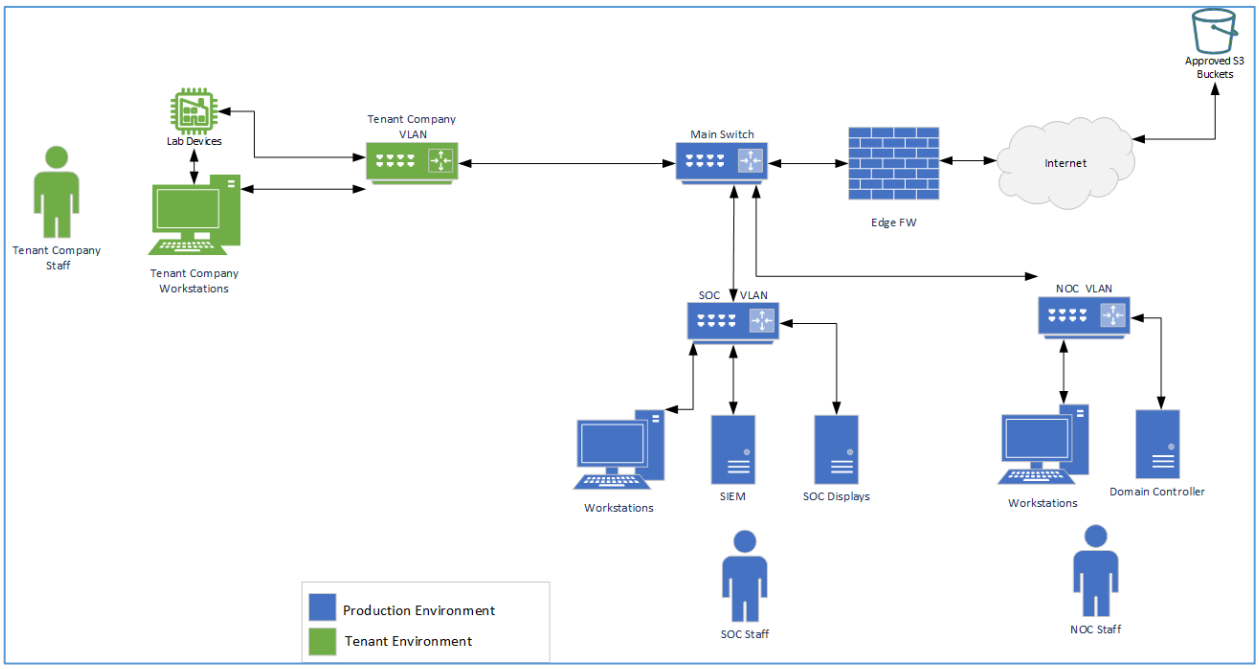


Figure 1 General Architecture Representation

4.1.2 Pilot-Specific Architecture

Building upon the representative environment, APL integrated a series of pilot-specific capabilities designed to support IOB development, visualization, and response automation. A dedicated network sensor was deployed to monitor a span port replicating production network traffic from the genomics tenant VLAN, enabling collection of packet-level and flow data used in behavior identification and validation.

A suite of CTI tools was installed to visualize and analyze IOB data, providing analysts with the ability to examine behavioral indicators, correlation logic, and relationships across multiple data sources. In parallel, a SOAR platform was integrated to execute the correlation workflows and detection analytics contained within the IOB bundles. These technologies collectively allowed APL to test and refine end-to-end IOB processes, from data ingestion and behavioral detection to automated response, prior to transitioning the configuration into the HudsonAlpha production environment. These additions to the architecture are illustrated in Figure 2.

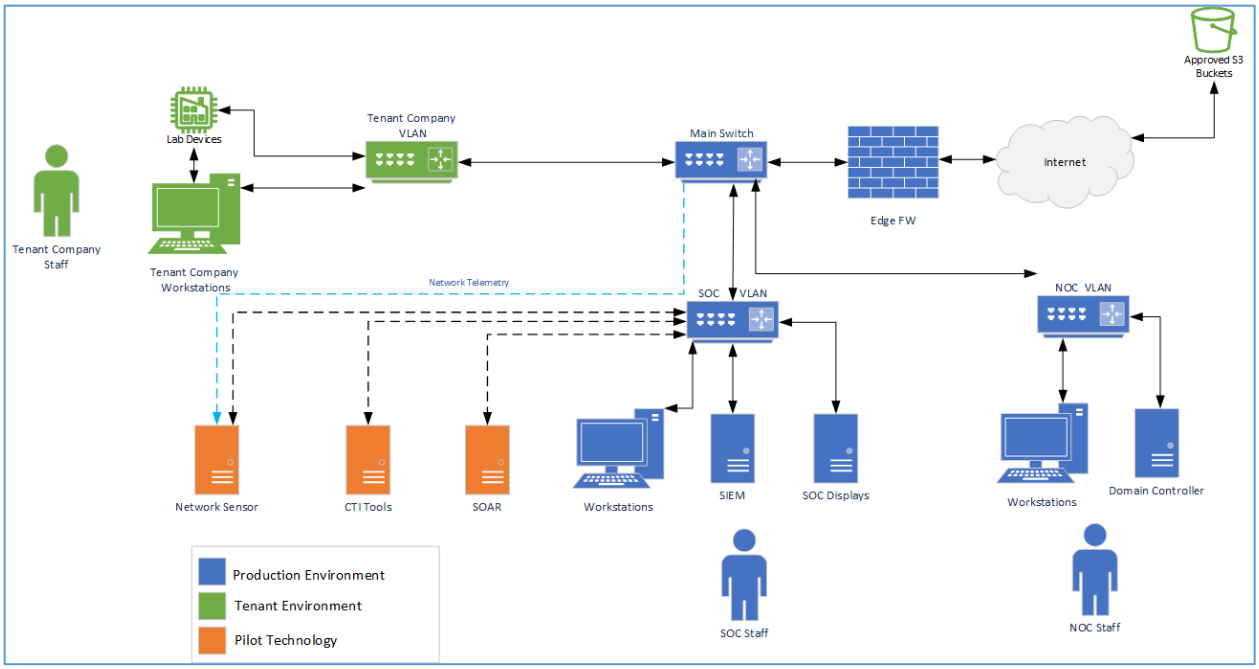


Figure 2 Technology Additions for IOB Pilot

4.2 Use Case Development

The Design Phase required the definition and implementation of two pilot use cases designed to evaluate the applicability of IOBs to real-world security operations within the HudsonAlpha environment. Each use case represented a distinct class of adversarial activity relevant to biotechnology operations, data theft and data integrity compromise, and provided a framework for developing corresponding behaviors, detections, and automated courses of action.

Both use cases were constructed and validated in APL's integration environment prior to deployment, allowing for controlled threat emulation, iterative analytic refinement, and validation of data collection pipelines. The use cases were then operationalized through IOB STIX bundles containing defined behaviors, detection logic, playbooks, and response workflows, each mapped to MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) tactics and techniques and supported by the pilot technology integrations.

4.2.1 Use Case 1: Data Theft

The first use case modeled an attacker compromising a tenant company operating within the HudsonAlpha campus to exfiltrate sensitive research data. The scenario demonstrated a full attack chain from initial compromise and credential theft through lateral movement and data exfiltration. This use case highlighted the risk of persistent access, misuse of legitimate cloud storage, and theft of proprietary FASTA/FASTQ datasets. Figure 3 provides an illustration of the threat activity found within Use Case 1.

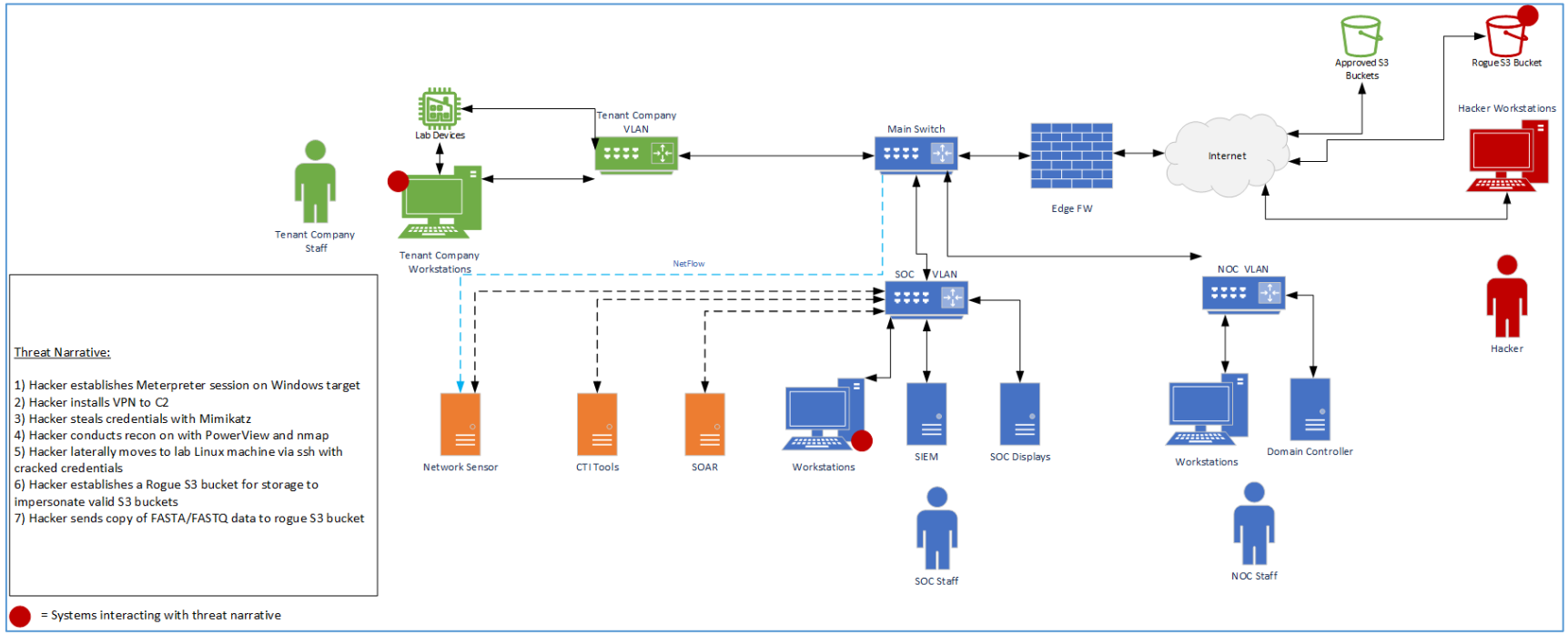


Figure 3 Use Case 1 Threat Activity

4.2.2 Use Case 1 Threat Narrative

The following narrative outlines the sequence of actions carried out by the simulated adversary during the data theft scenario. Each step corresponds to observable behaviors and associated detection opportunities within the emulated HudsonAlpha environment.

1. The attacker establishes a Meterpreter session on a Windows target system within the genomics tenant VLAN.
2. A VPN client is installed to maintain command and control connectivity with external infrastructure.
3. Credentials are stolen using Mimikatz, allowing lateral movement to other systems.
4. Reconnaissance is performed with PowerView and nmap to enumerate the internal network.
5. The attacker moves laterally to a Linux laboratory system using SSH and the compromised credentials. The target data are raw sequencing outputs (FASTA/FASTQ) generated by the campus's gene-sequencers and stored in the research data lake.
6. A rogue S3 bucket is created to impersonate a legitimate cloud storage repository.
7. Sensitive FASTA/FASTQ data is copied from the lab system and exfiltrated to the rogue bucket.

4.2.3 Use Case 1 Threat Behaviors

For Use Case 1, APL identified three primary behaviors, each tied to MITRE ATT&CK attack patterns:

Behavior Name	MITRE ATT&CK	Description
PowerView Enumeration	Discovery | T1018	A computer is utilizing PowerView PowerShell scripts to enumerate the network.
Lateral Movement via SSH	Lateral Movement | T1021.004	An SSH connection was established.
Establish Connection to Amazon S3 Bucket	Exfiltration | T1567.002	A computer established a connection to an Amazon S3 bucket.

4.2.4 Use Case 2: Data Integrity Compromise

The second use case focused on an adversary compromising a genetics lab environment to corrupt critical research data. This scenario was intended to illustrate the risks associated with unauthorized data manipulation and the difficulty of detecting attacks targeting research integrity rather than availability or theft. The attack chain included initial access via external remote services, lateral movement using VNC connections, and execution of malicious PowerShell scripts to modify or delete data. It also simulated attacker cleanup activities to reduce the likelihood of detection. Figure 4 provides an illustration of the threat activity found within Use Case 2.

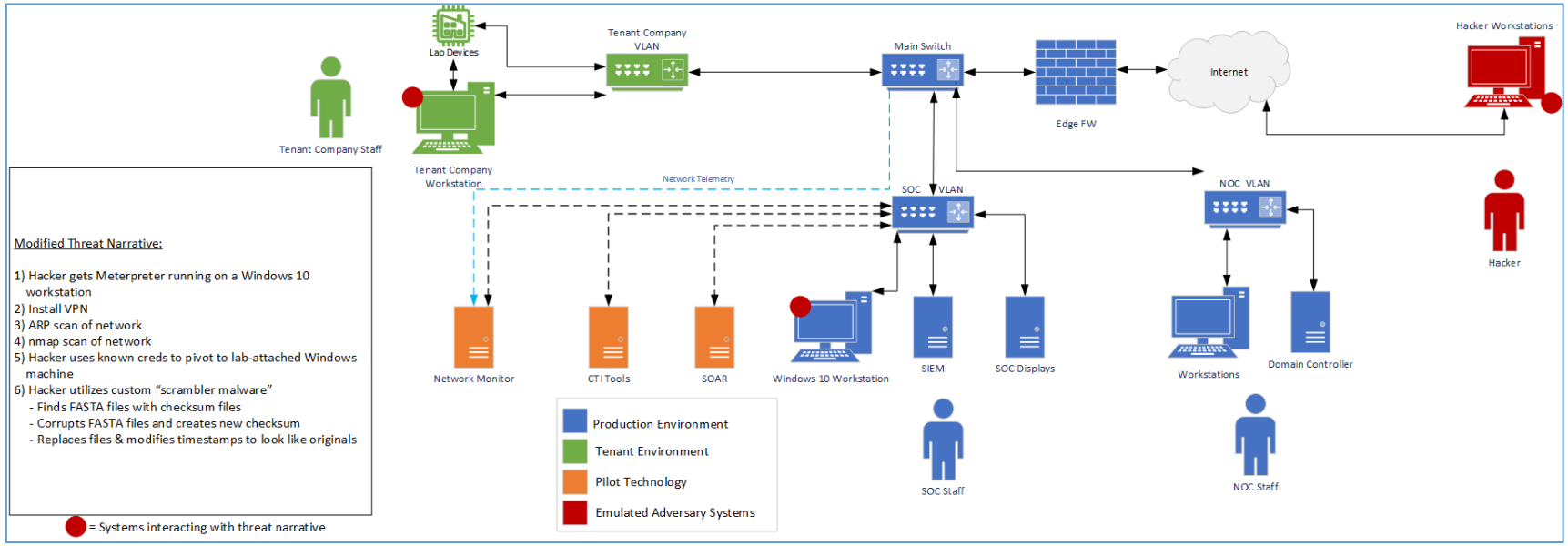


Figure 4 Use Case 2 Threat Activity

4.2.5 Use Case 2 Threat Narrative

The following narrative describes the adversary's actions during the data integrity compromise scenario. Each step represents a phase in the simulated attack chain, highlighting points of behavioral detection and opportunities for automated correlation within the pilot architecture.

- The adversary gains access through an initial Meterpreter session on a workstation.
- A persistent connection is established to maintain remote access to the genomics lab network through establishing a VPN.
- The adversary conducts scanning of the network.
- The adversary utilizes additional reconnaissance to identify targets.
- The adversary establishes a VNC connection to pivot to a lab-attached Windows machine.
- The adversary utilizes PowerShell to run "scrambler malware" to modify genomic data.

4.2.6 Use Case 2 Threat Behaviors

For Use Case 2, APL identified three primary behaviors, each tied to MITRE ATT&CK attack patterns:

Behavior Name	MITRE ATT&CK	Description
New Unknown External Connection Established	Persistence | T1133	A connection to an external IP address with an unrecognizable port was established.
New VNC Connection Established	Lateral Movement | T1021.005	A VNC connection was established.
PowerShell Script Execution	Execution | T1059.001	Execution of PowerShell scripts

4.3 Detection and Correlation Development

For each use case, APL developed a set of detection analytics and correlation workflows to include in the pilot IOB bundles. By operationalizing these analytics through the pilot technology integrations, APL enabled end-to-end testing of automated detections and behavior-based correlation within a representative environment prior to migration to HudsonAlpha's production systems.

4.3.1 Use Case 1 Detections

For the Data Theft scenario, three primary detections were implemented, each aligned to a specific adversary behavior identified from the emulated threat activity.

- PowerView Detection:

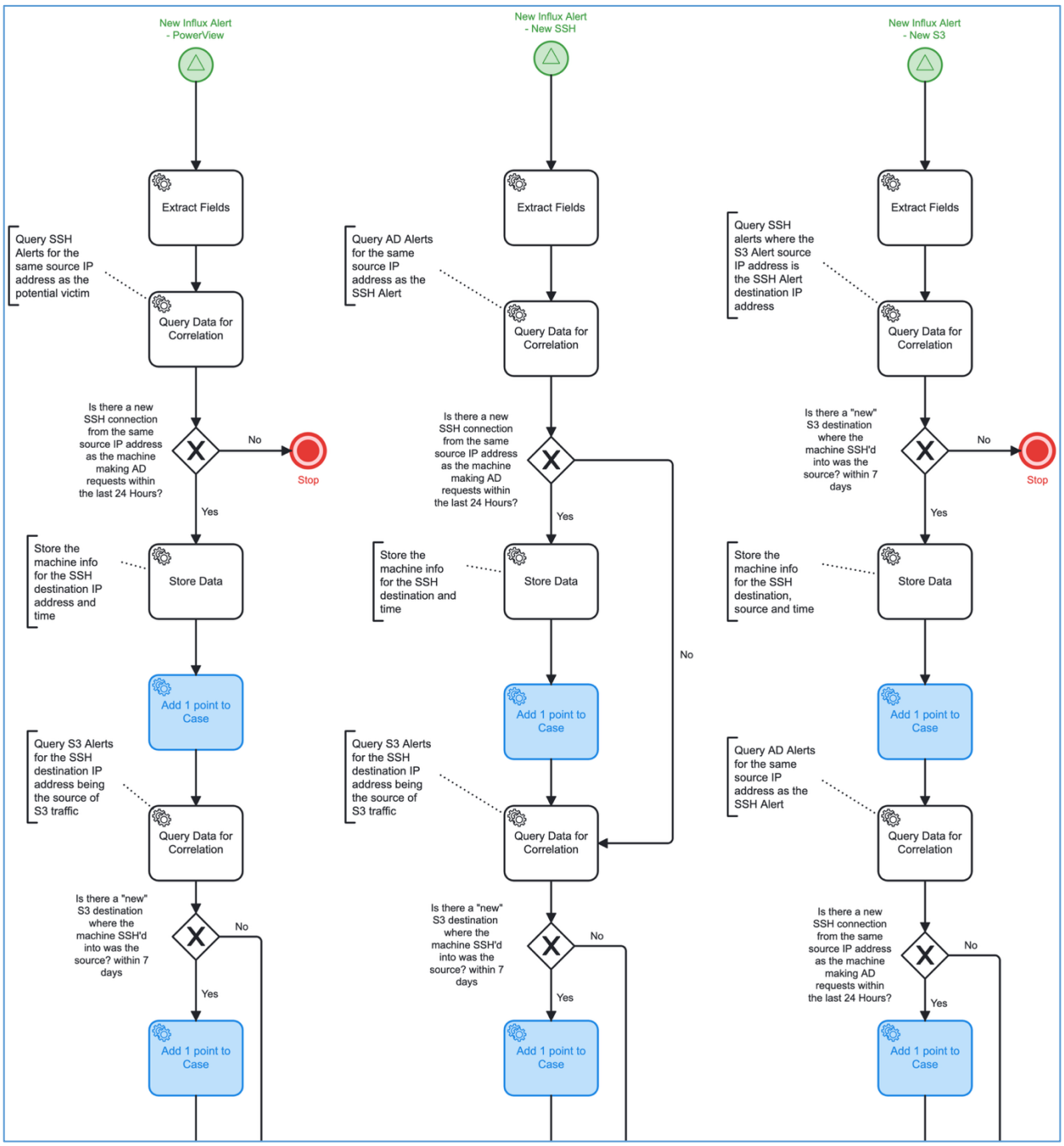
  - Implemented as a Sigma analytic designed to detect execution of PowerView or similar enumeration scripts.
  - The analytic monitors log sources from the network sensor for abnormal amounts of LDAP queries associated with the use of PowerView to enumerate a network.
- SSH Session Detection:
  - A Sigma-based analytic configured to flag creation of new SSH services or sessions across internal systems.
  - The analytic leverages network monitor data for any new internal source-destination pairings for SSH sessions.
- Establishing Connection to S3 Bucket Detection:
  - A network-based analytic created that monitors outbound traffic patterns consistent with Amazon S3 endpoints.
  - The analytic detects DNS lookups and network flows that involve a domain matching an S3 bucket pattern to alert on a connection to a new S3 bucket.

4.3.2 Use Case 1 Correlation

To enable automated scoring and triage of related detections, APL developed a course of action and security playbook titled "Correlate and Score Behaviors." This playbook, implemented in BPMN format, defines a procedure to automatically tally behavioral detections observed within a specified time window and produce a confidence score based on the number and sequence of correlated behaviors.

The workflow ingested alerts from the PowerView, SSH, and S3 exfiltration detections through the SOAR's interface. When two or more detections occurred within the same campaign context, the playbook incremented a cumulative score and generated an alert for operator review.

Figure 5 depicts the Use Case 1 correlation workflow implemented in BPMN format.



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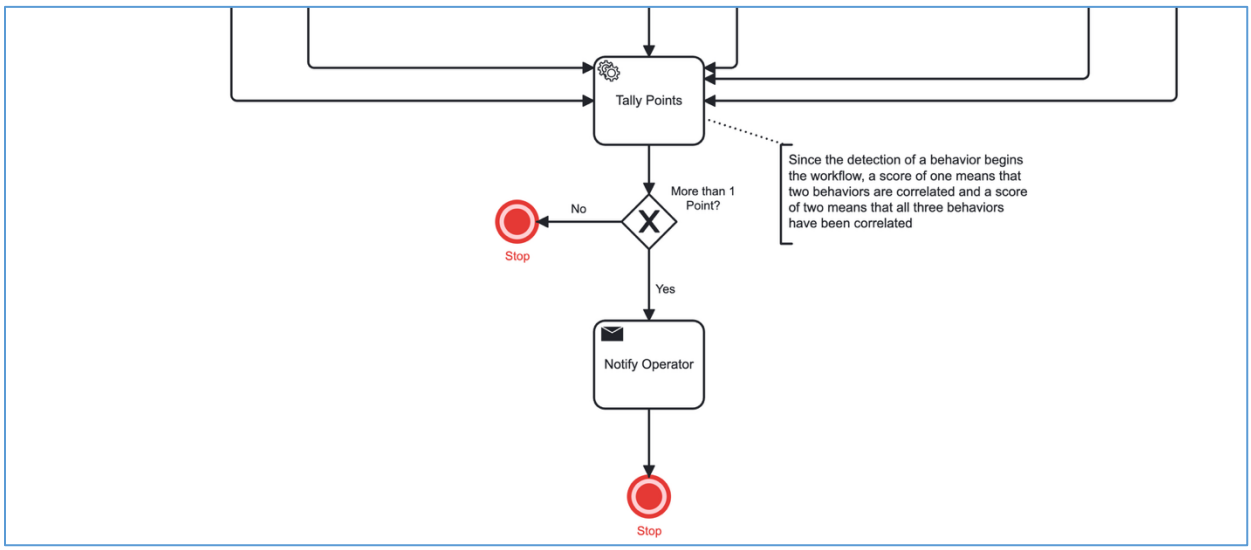


Figure 5 Use Case 1 Correlation Workflow

4.3.3 Use Case 2 Detections

For the Data Integrity Compromise scenario, three detection analytics were created to identify malicious connections, unauthorized remote control, and scripted execution.

- Unknown External Connection Establishment Detection:
  - A Sigma analytic configured to identify network connections to previously unseen external IP addresses.
- VNC Connection Establishment Detection:
  - A network monitoring analytic built to detect internal VNC sessions that may indicate lateral movement.
  - The analytic leveraged network monitor data to identify "rfb" protocol usage between lab hosts, flagging administrative or remote-control traffic inconsistent with normal operations.
- PowerShell Script Execution Detection:
  - A Sigma analytic designed to detect Windows PowerShell Event 4104 entries associated with execution of PowerShell scripts.

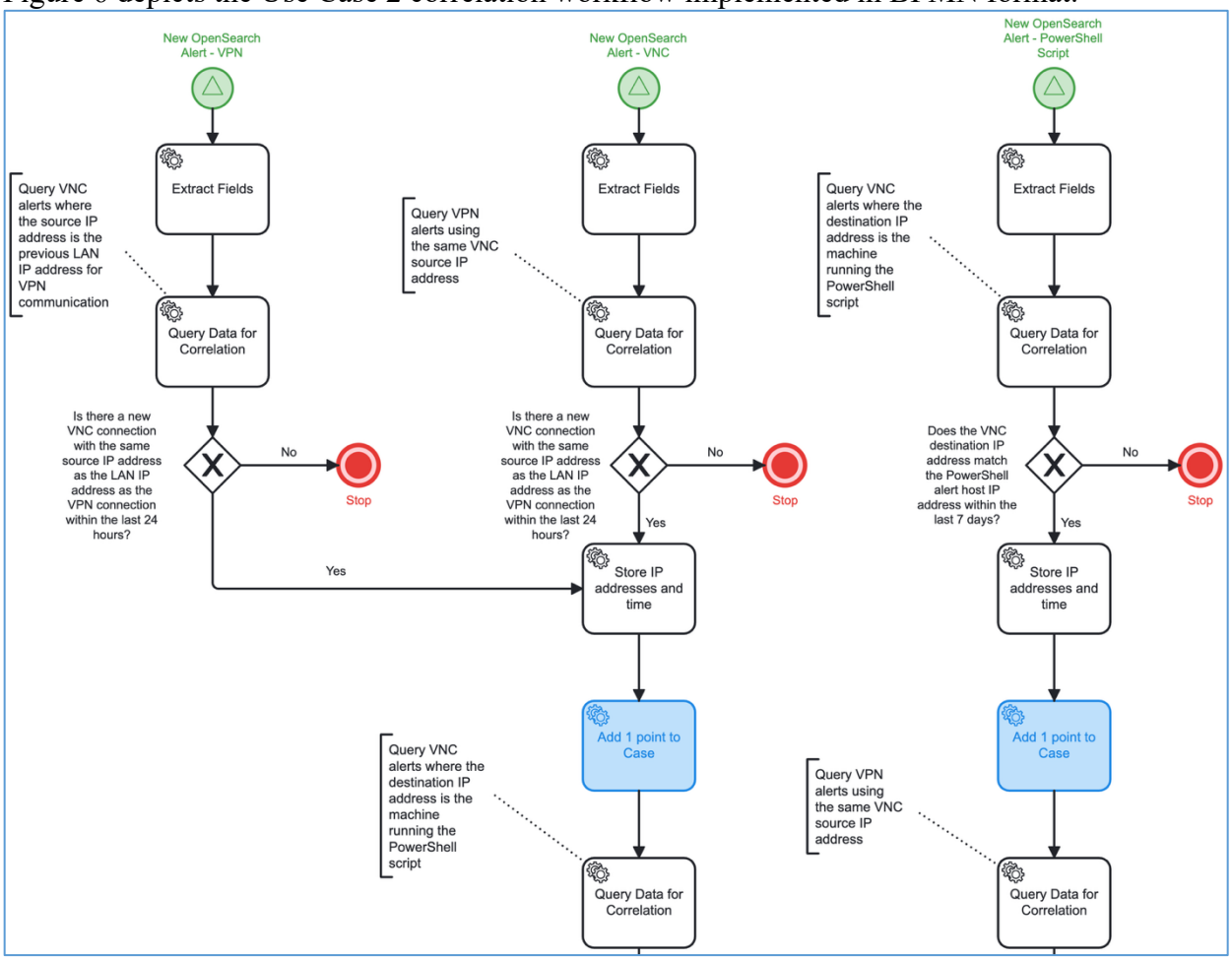
4.3.4 Use Case 2 Correlation

The correlation workflow for Use Case 2 extended the same "Correlate and Score Behaviors" logic used in the first scenario but adapted the analytic triggers and scoring weights to focus on research data integrity events.

The BPMN-based playbook correlated detections from the external connection, VNC, and PowerShell execution analytics. When multiple detections occurred within a defined time window, the workflow defines a procedure to elevate the incident priority, enrich the alert with contextual host and user information and issue notifications to SOC operators.

The correlation logic emphasized behavioral co-occurrence rather than alert count, because the correlation of alerts demonstrates the effectiveness of IOB-based detection for identifying subtle and coordinated attacks on laboratory systems.

Figure 6 depicts the Use Case 2 correlation workflow implemented in BPMN format.



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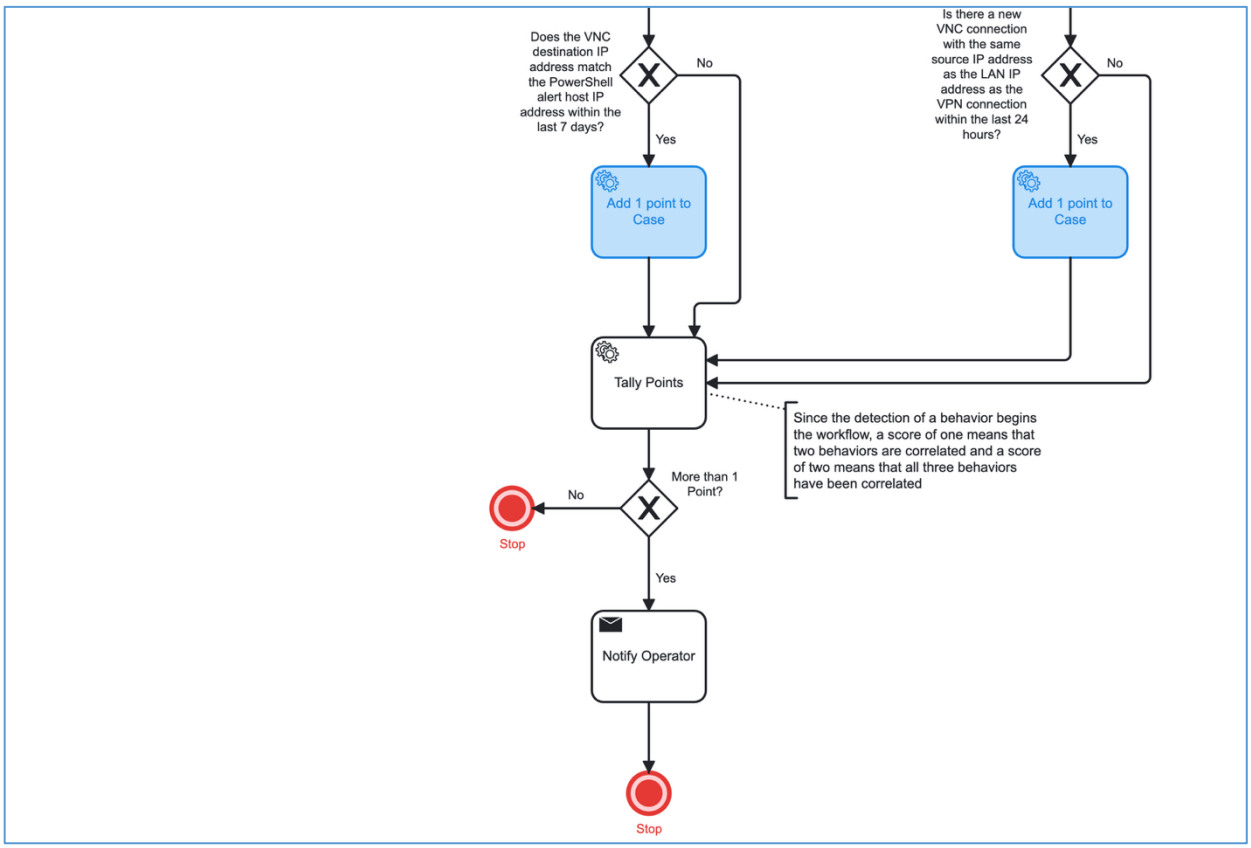

Figure 6 Use Case 2 Correlation Workflow

4.4 IOB Bundle Assembly

APL assembled the finalized IOB bundles using open-source tooling aligned with the OCA IOB extension schema. Each bundle contained the defined behaviors, detection analytics, courses of action, and correlation playbooks derived from the pilot use cases. The bundles were validated for STIX 2.1 compliance and structured to ensure interoperability with both commercial and open-source cybersecurity platforms. This process provided a portable, machine-readable representation of the detections and workflows developed during the Design Phase, enabling consistent deployment and evaluation across environments.

4.5 Integration and Testing

Following IOB bundle creation, APL conducted integration and testing activities within its controlled environment to validate analytic performance and automation reliability. The testing included live execution of emulated threat activity by Red Team operators to assess the responsiveness and accuracy of behavior-based detections. Automated correlation workflows within the environment successfully identified and linked multiple adversary behaviors, producing alerts in near real time. Across both pilot use cases, the time it took to perform correlation was within five minutes of an adversary behavior occurring on the network. These results demonstrated the operational feasibility of IOB-based analytics and provided baseline performance data for deployment preparation.

4.6 Migration to Pilot Deployment

Upon successful validation in the integration environment, APL worked closely with HudsonAlpha's SOC staff to migrate the pilot technologies and IOB bundles into production. The migration process included refinement of data collection pathways, verification of alert delivery mechanisms, and configuration adjustments to align with HudsonAlpha's existing workflows. APL provided consultation and technical support throughout the transition to ensure seamless integration and no disruption to active research or network operations. This phase concluded with the pilot capability fully deployed and ready for execution under operational monitoring conditions.

5. EXECUTION PHASE SUMMARY

Following integration and deployment of the IOB analytics and correlation workflows, the pilot entered a full operational execution period from August 19 through December 2. This phase measured real-world behavioral observations, correlation accuracy, false-positive suppression, and automated notification performance within the HudsonAlpha production environment.

5.1 Overall Pilot Performance

The deployed analytics and automation operated continuously and without disruption throughout the full three-month pilot window. Telemetry was ingested reliably, behavior detections fired as designed, and correlation workflows consistently executed in near real time.

A central objective of the pilot was to determine whether IOB-based detection could reduce the time required to identify coordinated threat activity. The results strongly validated that objective. Across the entire pilot, only high-confidence, fully correlated behaviors resulted in operator notifications, and these notifications were generated within five minutes of the final observed behavior.

Detection times using IOBs were significantly faster—and more accurate—than a manual processes, which could require days to weeks (up to 30 days) to identify similar patterns, if detected at all.

5.2 Use Case 1 – Data Theft Scenario

Use Case 1 evaluated behavioral detection and correlation for a multi-step data theft scenario consisting of three IOB-defined behaviors:

- PowerView Enumeration
- Lateral Movement via SSH
- Establish Connection to Amazon S3 Bucket

The detections for these behaviors were deployed in production beginning August 19, with continuous collection through December 2. Figure 7 provides a visualization of the behaviors and correlations observed for Use Case 1 in the Execution phase of the pilot.

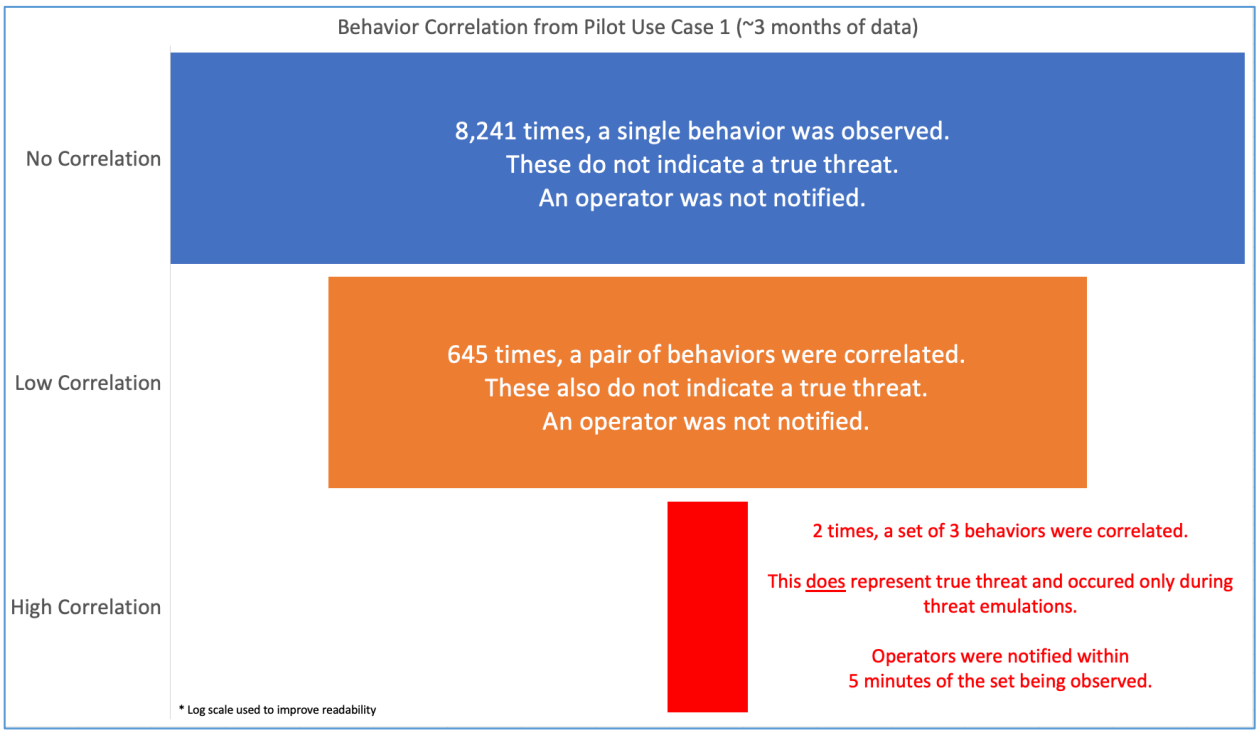

Figure 7 Overall Correlation Summary for Use Case 1

5.2.1 Observed Behaviors During Production

Across the pilot window, the automation recorded:

- 8,241 instances of single behaviors
  - These were logged but not shown to the operator, as individual behaviors are expected background noise and not indicative of malicious activity.
- 645 instances where two behaviors correlated
  - These partial matches were also logged but not escalated, demonstrating correct suppression of low-confidence events.
- 2 instances of full three-behavior correlation
  - These occurred only during intentional threat-emulation exercises.
  - Both instances were immediately classified as malicious and were sent to the operator within five minutes of occurrence.

This distribution demonstrates a core strength of IOB correlation. Thousands of benign or low-confidence observations were automatically suppressed, while the only true malicious sequences were identified and escalated with high precision. Figure 8 provides a daily distribution of the observed behaviors and correlations for Use Case 1 and Figure 9 provides the daily summary of high correlations which resulted in an operator notification.

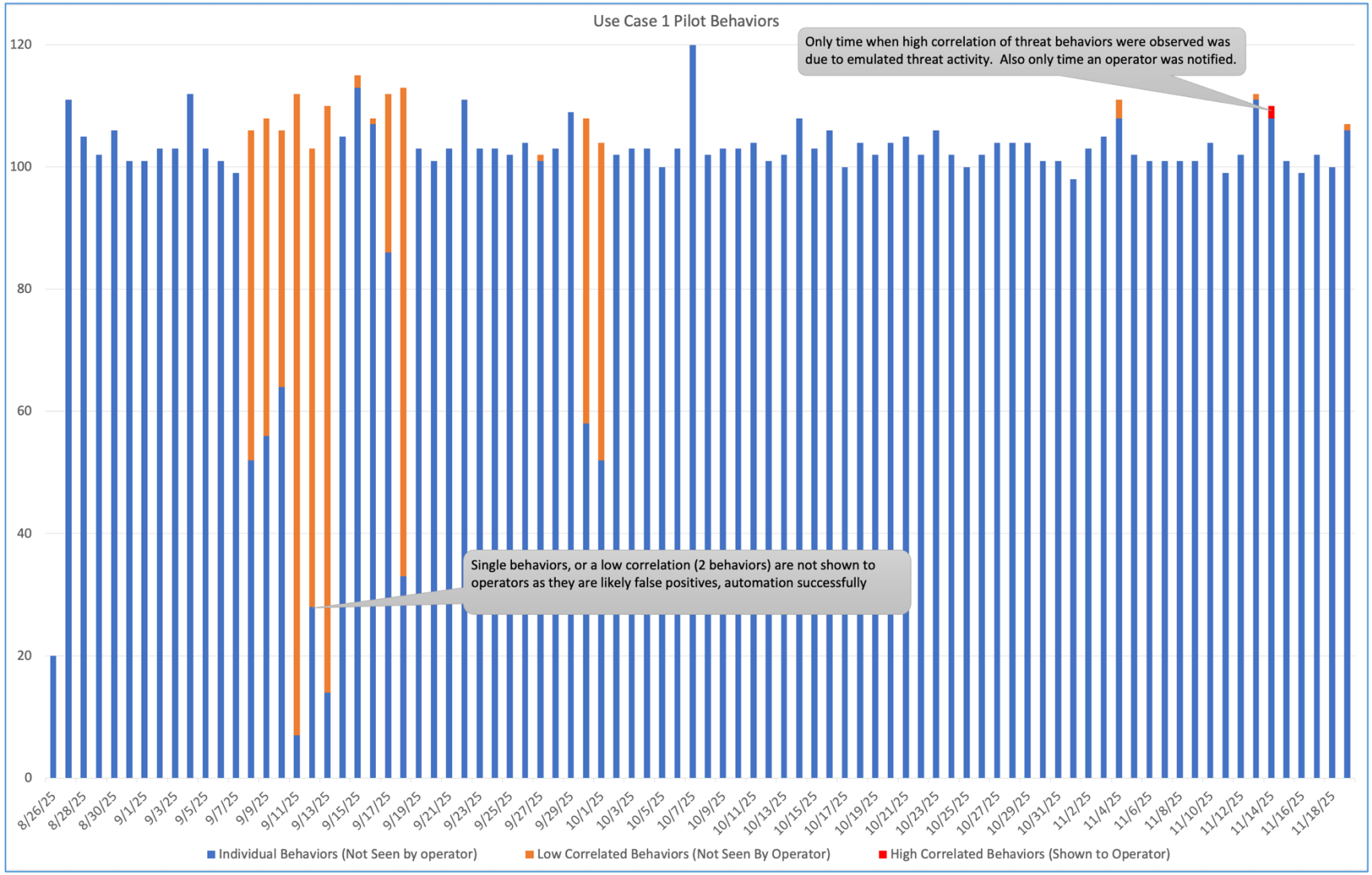

Figure 8 Daily Summary of Use Case 1 Behavior Correlations

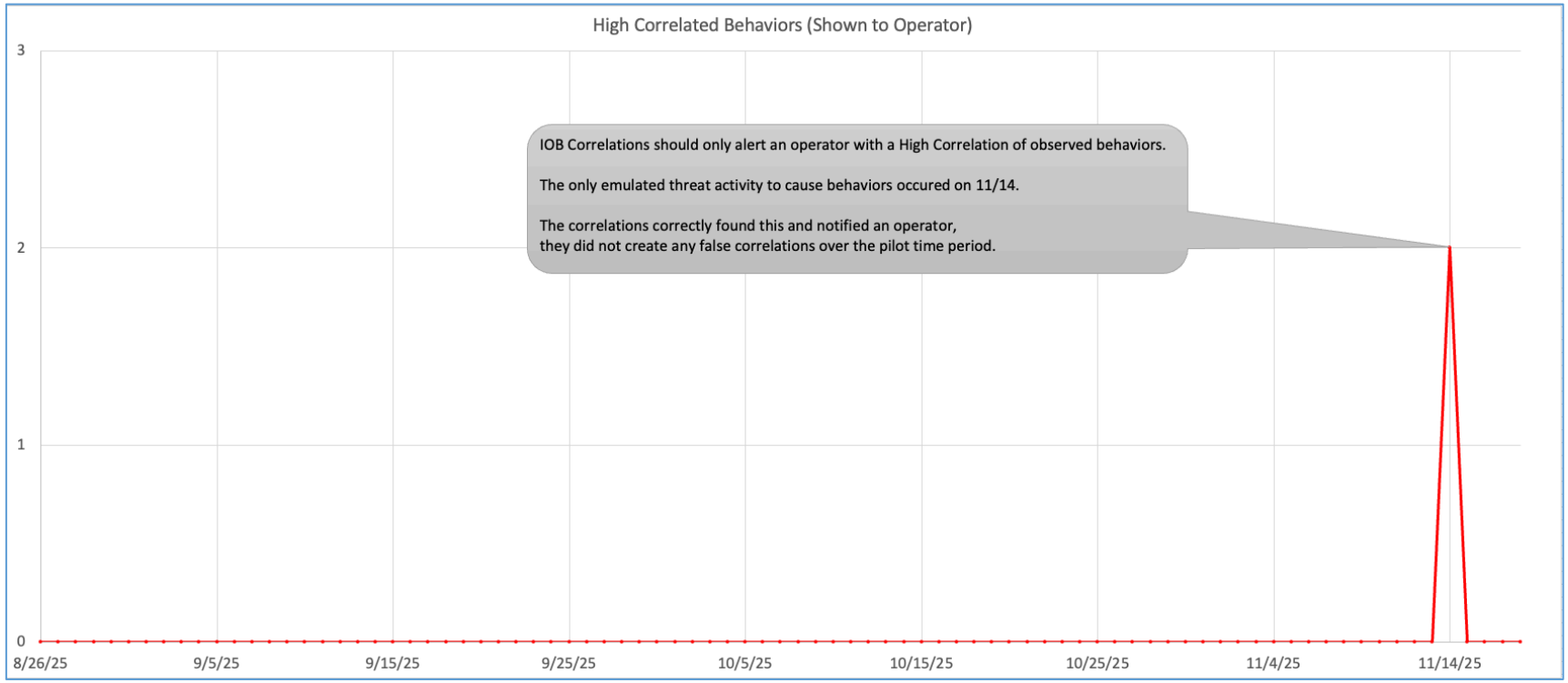

Figure 9 Daily Summary of Use Case 1 Operator Notifications

5.3 Use Case 2 – Data Integrity Compromise

Use Case 2 evaluated IOB applied to an adversary scenario focused on research data integrity rather than data theft. This use case exercised detection analytics and a correlation workflow on endpoint-centric and host-interaction behaviors observed within HudsonAlpha's laboratory environment. The goal was to assess whether IOBs could identify subtle adversary activity designed to corrupt genetic data while avoiding detection.

This scenario represented a realistic and high-impact threat vector for biotechnology organizations, where data manipulation, even without exfiltration, can compromise research validity, disrupt experiments, and introduce long-term scientific or operational harm.

Use Case 2 evaluated behavioral detection and correlation for a multi-step data theft scenario consisting of three IOB-defined behaviors:

- New VPN Connection Established
- New VNC Connection Established
- PowerShell Script Execution

The detections for these behaviors were deployed in production beginning August 19, with continuous collection through December 2. Figure 10 provides a visualization of the behaviors and correlations observed for Use Case 2 in the Execution phase of the pilot.

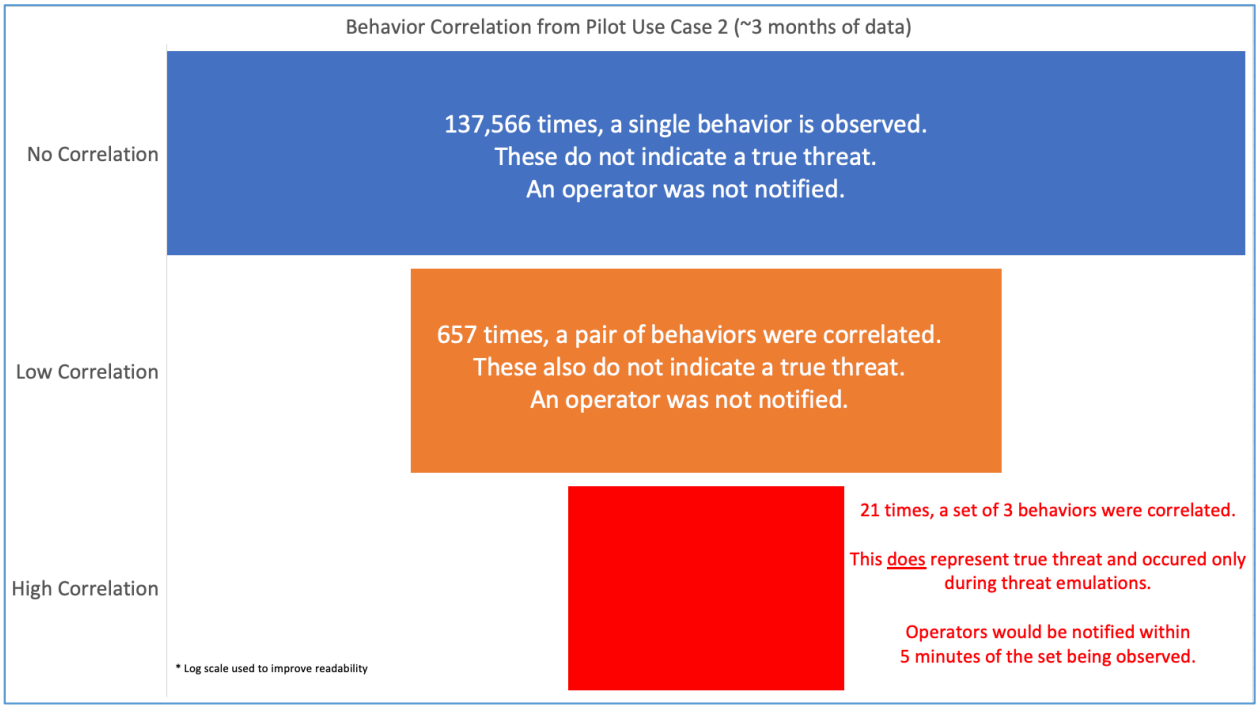

Figure 10 Overall Correlation Summary for Use Case 2

5.3.1 Observed Behaviors within Lab Environment

Across the pilot window, the automation recorded:

- 137,566 instances of single behaviors
  - These were logged but not shown to the operator, as individual behaviors are expected background noise and not indicative of malicious activity.
- 657 instances where two behaviors correlated
  - These partial matches were also logged but not escalated, demonstrating correct suppression of low-confidence events.
  - This number is underreported due to a SIEM configuration that only returned the first ten alerts for a behavior
- 21 instances of full three-behavior correlation
  - These occurred only during intentional threat-emulation exercises.
  - All instances were immediately classified as malicious and a notification would be sent to the operator within two minutes of occurrence. (Note: to avoid disrupting actual security operations, the notifications from the lab environment were not implemented during the pilot)

This distribution demonstrates a core strength of IOB correlation. Thousands of benign or low-confidence observations were automatically suppressed, while the only true malicious sequences were identified and escalated with high precision. Figure 11 provides a daily distribution of the observed behaviors and correlations for Use Case 2 and Figure 12 provides the daily summary of high correlations which would result in an operator notification.

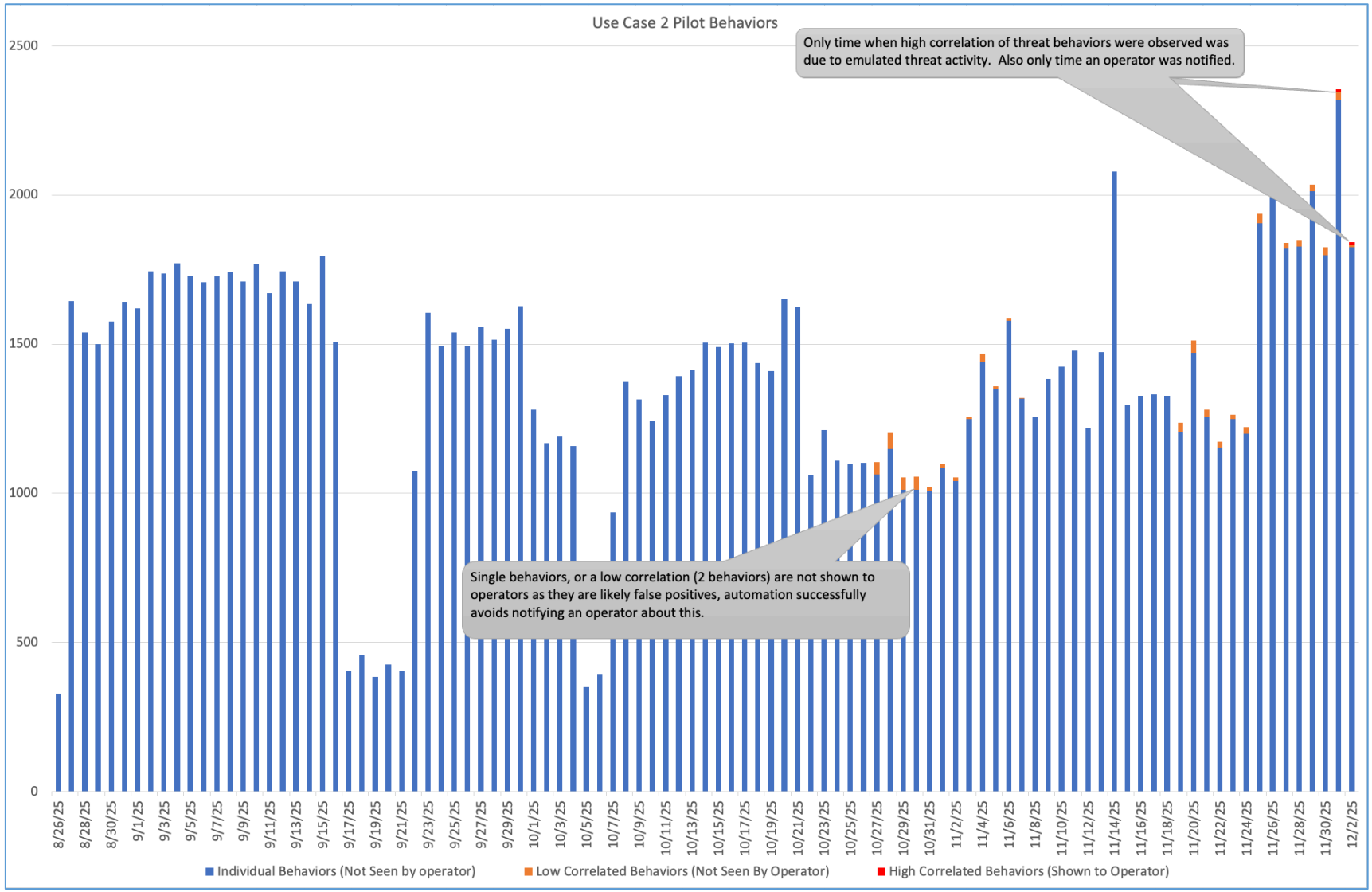

Figure 11 Daily Summary of Use Case 2 Behavior Correlations

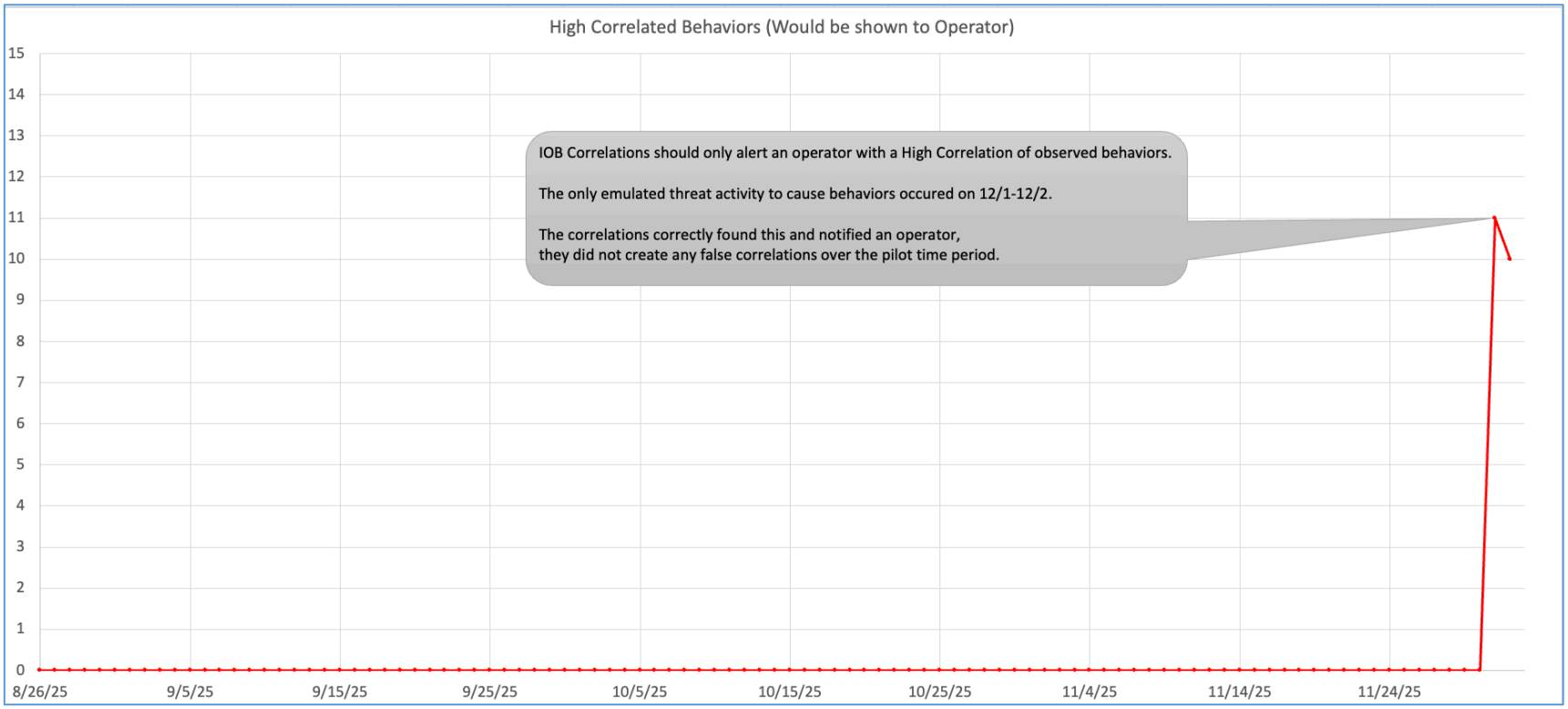

Figure 12 Daily Summary of Use Case 2 Expected Operator Notifications

5.4 Operational Validation

Across ~90 days of continuous operation:

- No false-positive high-correlation events occurred.
- Only the intended adversary emulation events triggered operator alerts.
- The automation performed with minimal interruptions and no misclassification.
- Alert fatigue was eliminated—operators received exactly two alerts, both valid.

These results reflect a high-accuracy, low-noise analytic pipeline appropriate for operational deployment within a production SOC environment.

6. CONCLUSION

The HudsonAlpha IOB Operational Pilot provided a clear demonstration that behavior-based cyber analytics can function effectively in a live production environment. Throughout approximately ninety days of continuous execution, the pilot systems operated reliably, processed telemetry as expected, and produced high-confidence alerts only when emulated adversary activity occurred. There were no false positives, no misclassifications, and minimal interruptions to operational workflows. The pilot's success was measured against realistic workloads, including continuous gene-sequencer data flows, demonstrating that the IOB framework can protect both enterprise services and core genomic research pipelines.

Both use cases supported this conclusion. In the data theft scenario, individual behaviors were common but only coordinated malicious sequences resulted in correlation events. In the data integrity compromise scenario, the system successfully identified the full set of intended lab-oriented behaviors while suppressing a large volume of benign activity. In both environments, the pilot demonstrated consistent behavior and strong alignment with operational needs.

These results illustrate the value of adopting behavioral correlation as a complement to existing cyber defense approaches. The pilot confirmed that IOBs can enhance analyst awareness, reduce noise, and support timely identification of multi-step adversary activity. The approach is suitable for organizations with diverse environments, including research campuses and biotechnology operators, and can be scaled to additional sectors of critical infrastructure.

The collaboration among APL, CISA, OCA, and HudsonAlpha was instrumental in validating both the technical feasibility and operational utility of the capability. The findings provide a strong foundation for wider deployment of IOB-based analytics and standards. They also demonstrate a realistic path for transitioning emerging cyber defense methods into production settings where accuracy, reliability, and operational fit are essential.